



**HEAT BALANCE LAB QUESTIONS:**

1. Why does the Core  $\Delta T$  ( $T_4 - T_2$ ) not always agree with the Heat Exchanger  $\Delta T$  ( $T_5 - T_6$ )?
2. When power was increased, why was there a lag before temperature increased on  $T_4$ ?
3. When power was increased, why was there a lag between the temperature increase indicated on  $T_4$  and  $T_5$ ?
4. Does the temperature measured by  $T_2$  accurately represent the temperature of the entire reactor pool?
5. What was the calibration factor for the  $^{16}\text{N}$  channel as measured in lab? Is this conservative or non-conservative?
6. When the Cold Primary Return temperature  $T_3$  (or  $T_6$ ) is too high relative to the system setpoint (typically set at  $90^\circ\text{F}$ ):
  - a. What physical parameter does the temperature control system change to cause the temperature to decrease?
  - b. Why doesn't  $T_6$  decrease immediately (i.e. why is there a lag) after the control system makes the change in (a) above?